Curvigram Explained

A non-technical guide for archaeoastronomical research

1. Basic idea

A histogram places observations into bins. A curvigram represents every observation with a smooth local kernel and then sums the contributions into a continuous curve.

In this workshop the curvigram is deliberately transparent: participants can see the kernel formula, bandwidth rule, summation and area normalisation in the R script.

2. One observation, one kernel

For an observation at value `x_i`, calculate:

```
u = (x - x_i) / h_i
```

where `h_i` is that observation's bandwidth.

The observation contributes:

```
K(u) / h_i
```

to the curve.

3. Epanechnikov kernel

Workshop baseline:

```
K(u) = 0.75 x (1 - u^2), when |u| <= 1
K(u) = 0,                otherwise
```

It has finite support: beyond the bandwidth its contribution is exactly zero.

4. Gaussian kernel

Sensitivity comparison:

```
K(u) = exp(-u^2 / 2) / sqrt(2*pi)
```

Its tails decrease smoothly rather than ending at a finite boundary.

5. Bandwidth

The bandwidth controls the width of every local contribution.

Workshop rule:

```
h_i = multiplier x uncertainty_i
```

- smaller multiplier -> sharper, more local structure;
- larger multiplier -> smoother curve, greater merging of nearby features.

The multiplier is a methodological choice, not a decorative graph setting.

6. Meaning of uncertainty

`uncertainty_deg` is the **reported plus/minus measurement uncertainty**.

The workshop does **not** automatically assume that it is one standard deviation.

If your measurement protocol gives it another statistical meaning, document that meaning explicitly.

7. Default area normalisation

The individual kernels are summed and the displayed curve is divided by its numerical area so that:

```
total displayed area = 1
```

This makes the vertical scale a density-like shape that is convenient for comparing distributions.

8. Linear versus circular geometry

The main workshop curvigram uses **astronomical declination** as a linear coordinate over the analysed range.

If you smooth azimuths, 359 degrees and 1 degree must be treated as neighbours. Circular boundary handling is then essential.

9. Frozen workshop baseline

```
Variable:          declination_deg
Geometry:          linear
Kernel:            Epanechnikov
Bandwidth rule:    2 x uncertainty_deg
Vertical scale:    area = 1
Range:             -45 to +45 degrees
Grid step:         0.05 degrees
```

10. Required sensitivity workflow

Compare:

```
Epanechnikov x 1
Epanechnikov x 2   baseline
Epanechnikov x 3
Gaussian x 2
```

Ask:

1. Does the major feature remain in approximately the same place?
2. Does it remain clearly visible?
3. Do smaller features merge, flatten or disappear?
4. Would the substantive interpretation change?

11. What a peak means

A peak indicates relatively high combined density in that coordinate region.

Possible explanations include:

- a genuine preferred orientation;
- a cultural convention;
- an astronomical target;
- topographic or architectural constraints;
- multiple practices;
- sampling structure;
- a feature amplified by smoothing choices.

The graph alone cannot decide between these explanations.

12. What a peak does not prove

A peak close to a solar, lunar or stellar value is not by itself proof of intentional astronomical alignment.

Interpretation should consider:

- archaeological context;
- chronology;
- landscape/topography;
- horizon visibility;
- measurement uncertainty;
- plausible alternative targets;
- possible post-hoc target selection;
- robustness to reasonable analytical specifications.

13. Reporting checklist

Report at least:

- coordinate analysed;
- linear/circular treatment;
- sample size;
- meaning of the uncertainty variable;
- kernel;
- bandwidth rule;
- multiplier;
- vertical scaling;
- calculation range;
- calculation step;
- sensitivity cases;
- software used.

14. Take-home message

A curvigram is not a photograph of the true distribution. It is a transparent representation produced from observations plus explicit smoothing choices. Its evidential value increases when those choices are reported and tested for robustness.